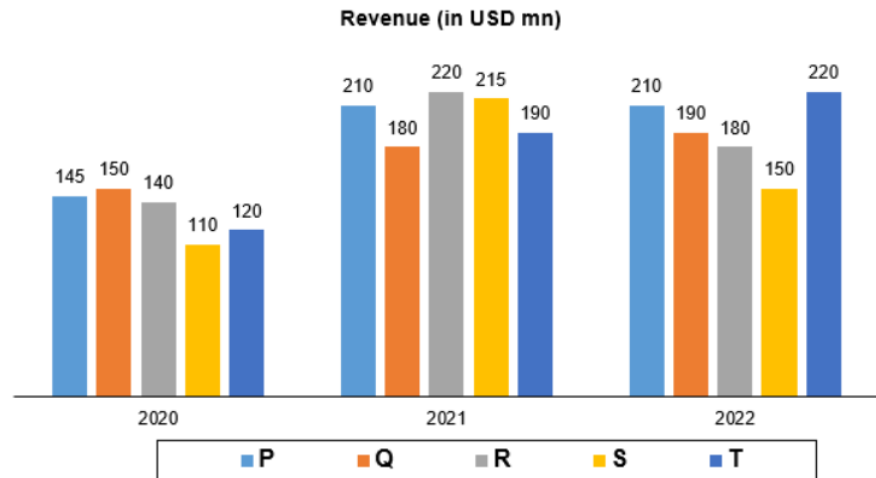


DIRECTIONS for questions 1 to 5: Answer the questions on the basis of the information given below.

The graph below provides the revenues of five companies, P, Q, R, S and T, for each of the three years from 2020 to 2022. In each of 2021 and 2022, the expenditure of each of the five companies increased by 20% as compared to that in the previous year.



It is also known that

- i. the profit of a company is defined as Revenue – Expenditure
- ii. the profit percentage of a company is defined as $\frac{\text{Profit}}{\text{Expenditure}} \times 100$
- iii. in 2020, the company with the highest profit percentage incurred an expenditure of USD 80 mn.
- iv. in 2021, the company with the highest profit percentage incurred an expenditure of USD 102 mn.
- v. in 2022, the company with the highest profit percentage incurred an expenditure of USD 108 mn.
- vi. in 2021, one of the companies incurred an expenditure of USD 98 mn, while another company incurred an expenditure of USD 100 mn.

Q1. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.

What is the expenditure (in USD mn) of Q in 2022?

- ☐ a) 115.2
- ☐ b) 122.4
- ☐ c) 120
- ☐ d) 117.6

Q2. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.

Which of the five companies incurred the highest expenditure in 2020?

☐ a) P

☐ b) Q

☐ c) R

☐ d) S

Q3. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.

What is the second highest profit percentage of any of the five companies in 2021?

- ☐ a) 115.68%
- ☐ b) 115%
- ☐ c) 114.29%
- ☐ d) 111.11%

Q4. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.

In 2020, how many of the five companies had a higher profit than T?

- ☐ a) 0
- ☐ b) 1
- ☐ c) 2
- ☐ d) More than 2

Q5. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.

What is the profit of the company with the third highest profit percentage in 2022?

- ☐ a) USD 74.8 mn
- ☐ b) USD 64.8 mn
- ☐ c) USD 62.4 mn
- ☐ d) USD 92.4 mn

DIRECTIONS for questions 6 to 10: Answer the questions on the basis of the information given below.

Four persons – Kishan, Ratan, Latif and Pavan – work in a factory and on each day, they have to fill water in nine containers which are arranged in three rows and three columns on the factory floor. Each container has a capacity of 100 litres. The nine containers are numbered as shown in the following figure:

1	2	3
4	5	6
7	8	9

However, some of the nine containers are defective, i.e., they have a hole in the bottom, because of which any quantity of water poured into these containers will be drained out fully in not more than 10 minutes. The four persons fill the containers on each day in the following manner:

- At exactly 9:00 AM, Kishan fills each of containers 1, 2, 4 and 9 with 20 litres of water.
- At exactly 11:00 AM, Ratan fills all the containers to the immediate right of the containers which are not empty. Ratan also fills any container with exactly 20 litres of water.
- At exactly 1:00 PM, Latif fills all the containers which are immediately below the containers which are not empty. Latif also fills any container with exactly 20 litres of water.
- At exactly 3:00 PM, Pavan fills all the containers which are to the immediate left of the containers which are not empty. Pavan also fills any container with exactly 20 litres of water.

At 6:00 PM, all the water in the containers are used for cleaning the machines in the factory. Until 6:00 PM, the water in the containers are not consumed in any way.

Q6. DIRECTIONS for questions 6 to 10: Select the correct alternative from the given choices.

If, at 5:00 PM, none of the containers have more than 40 litres of water, what is the minimum number of defective containers?

- ☐ a) 1
- ☐ b) 2
- ☐ c) 3
- ☐ d) 4

Q7. DIRECTIONS *for questions 6 to 10:* Select the correct alternative from the given choices.

If container 8 was not filled by any of the four persons during the day, which of the following containers must definitely be defective

- ☐ a) Container 4
- ☐ b) Container 5
- ☐ c) Container 9
- ☐ d) More than one of the above

Q8. DIRECTIONS *for questions 6 to 10:* Select the correct alternative from the given choices.

If, at 2:00 PM, each of the non-defective containers has at most 20 litres of water, what is the minimum number of containers that must be defective?

☐ a) 2

☐ b) 1

☐ c) 3

☐ d) 4

Q9. DIRECTIONS *for questions 6 to 10:* Select the correct alternative from the given choices.

What is the maximum number of non-defective containers that would have been empty at 5:00 PM?

☐ a) 2

☐ b) 3

☐ c) 4

☐ d) 5

Q10. DIRECTIONS *for questions 6 to 10:* Select the correct alternative from the given choices.

If at 5:00 PM all the non-defective containers have exactly 40 litres of water, what is the minimum number of containers that must be defective?

☐ a) 3

☐ b) 4

☐ c) 5

☐ d) 6

DIRECTIONS for questions 11 to 15: Answer the questions on the basis of the information given below.

In a college, there are exactly 250 students and each student can speak at least one language among English, Spanish and French. Also, a certain number of these 250 students wanted to learn one or more languages – among English, Spanish and French – that they cannot speak.

It is also known that

- i. the numbers of students who can speak only English, only Spanish and only French are in the ratio 2 : 3 : 4.
- ii. exactly 20 students can speak all the three languages.
- iii. the number of students who want to learn only French is twice the number of students who can speak only English and Spanish.
- iv. the number of students who can speak only one language is thrice the number of students who want to learn English and Spanish.
- v. the number of students who can speak exactly two languages is twice the number of students who want to learn only English.
- vi. among the students who can speak exactly two languages, the number of students who can speak English, Spanish and French are in the ratio 2 : 1 : 2.

Q11. DIRECTIONS for questions 11 to 14: Type in your answer in the input box provided below the question.
How many students speak only English?

Q12. DIRECTIONS *for questions 11 to 14:* Type in your answer in the input box provided below the question.

What is the number of students who want to learn only French?

Q13. DIRECTIONS *for questions 11 to 14:* Type in your answer in the input box provided below the question.

The number of students who want to learn only Spanish is at most

Q14. DIRECTIONS *for questions 11 to 14:* Type in your answer in the input box provided below the question.

What is the maximum number of students who can speak French and want to learn English?

Q15. DIRECTIONS *for question 15:* Select the correct alternative from the given choices.

The number of students who want to learn both English and French is at most

- ☐ a) 45
- ☐ b) 55
- ☐ c) 25
- ☐ d) None of the above

DIRECTIONS for questions 16 to 20: Answer the questions on the basis of the information given below.

Dr. Bhairavnath runs a clinic in a small town. During a certain week, from Day 1 to Day 7, a certain number of patients visited his clinic for diagnosis. Each patient was suffering from exactly one ailment among headache, stomach ache, cough, cold and fever. Further, each patient visited the clinic exactly once during the week.

It is known that, among the patients who visited his clinic,

- i. on each of the seven days, the number of patients suffering from stomach ache was twice the number of patients suffering from fever.
- ii. on each of the seven days, the number of patients suffering from cold was five more than the number of patients suffering from headache.
- iii. on each of the seven days, the number of patients suffering from headache was ten more than the number of patients suffering from stomach ache.
- iv. on each of the seven days, the number of patients suffering from cough was greater than that of those suffering from any other ailment.
- v. for exactly one ailment, the number of patients suffering from that ailment was the same for each of the seven days.

Also, the total number of patients who visited his clinic during the week was 560. On each of Day 1 and Day 2, exactly 60 patients visited his clinic, while on each day from Day 3 to Day 7, an equal number of patients visited his clinic.

Q16. DIRECTIONS for question 16: Select the correct alternative from the given choices.

How many patients suffering from cold visited the clinic on Day 4?

- ☐ a) 25
- ☐ b) 23
- ☐ c) 21
- ☐ d) 17

Q17. DIRECTIONS *for questions 17 and 18:* Type in your answer in the input box provided below the question.

What is the total number of patients suffering from headache who visited the clinic during the week?

Q18. DIRECTIONS *for questions 17 and 18:* Type in your answer in the input box provided below the question.

What is the difference between the number of patients suffering from stomach ache who visited the clinic on Day 2 and the number of patients suffering from headache who visited the clinic on Day 7?

Q19. DIRECTIONS *for questions 19 and 20:* Select the correct alternative from the given choices.

Among the patients who visited the clinic during the week, what is the sum of the total number of patients suffering from cold and the total number of patients suffering from fever?

- ☐ a) 186
- ☐ b) 168
- ☐ c) 175
- ☐ d) 170

Q20. DIRECTIONS *for questions 19 and 20:* Select the correct alternative from the given choices.

Among the patients who visited the clinic during the week, the difference between the total number of patients suffering from ailment X and the total number of patients suffering from ailment Y is the highest. Which of the following pairs of ailments can represent X and Y, respectively?

- ☐ a) Cough and stomach ache
- ☐ b) Cold and stomach ache
- ☐ c) Cold and fever
- ☐ d) Cough and fever